

Comprehensive setup for Temperature, Humidity, and gas sensor fusion for Indoor Air Quality (IAQ)

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Problem Statement & objectives



- People spend most of their time indoors
- Poor indoor air quality affects health and productivity

- Traditional systems are expensive and lack real-time monitoring
- Proposed a low-cost IoT-based monitoring system



- Design a real-time multi-parameter indoor air quality monitoring system
- Calculate AQI using India-standard breakpoints for PM2.5, PM10, CO₂ and TVOC

- Transmit live sensor data to cloud via MQTT over 4G LTE network
- View over over a dashbord anytime and anywhere

AQI Calculation

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}} (C_p - BP_{Lo}) + I_{Lo}$$

I_p = Sub index of each pollutant

C_p = The measured concentration of a pollutant

BP_{Hi} (Breakpoint High) = The upper concentration breakpoint of C_P

BP_{Lo} (Breakpoint Low) = The lower concentration breakpoint of C_P

I_{Hi} (Index High) = The AQI value corresponding to BPH.

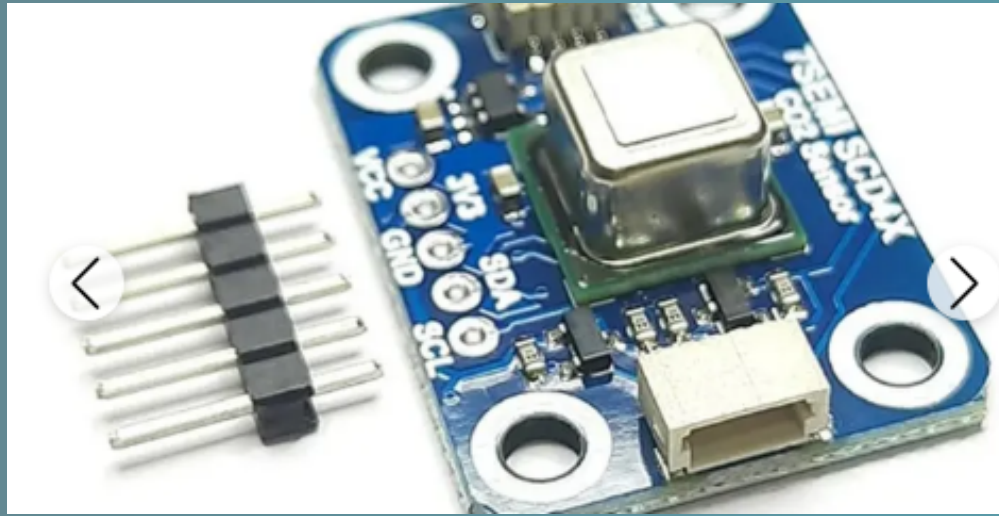
I_{Lo} (Index Low) = The AQI value corresponding to BPL

$$\text{AQI} = \max(I_p)$$

AQI calculations and Concentration breakpoints

AQI Range	Category	PM2.5	PM10	CO ₂ (ppm)	TVOC
0 – 50	Good	0 – 30	0 – 50	0 – 600	0 – 65
51 – 100	Satisfactory	31 – 60	51 – 100	601 – 1000	66 – 220
101 – 200	Mod. Polluted	61 – 90	101 – 250	1001 – 1500	221 – 660
201 – 300	Poor	91 – 120	251 – 350	1501 – 2000	661 – 2200
301 – 400	Very Poor	121 – 250	351 – 430	2001 – 5000	2201 – 5500
401 – 500	Severe	> 250	> 430	> 5000	> 5500

Hardware Components



SDC4x
CO2 , Temp , Humidity

I2C SDA = 2 , SCL = 1



ENS160
TVOC+eCO2

I2C SDA = 2 , SCL = 1

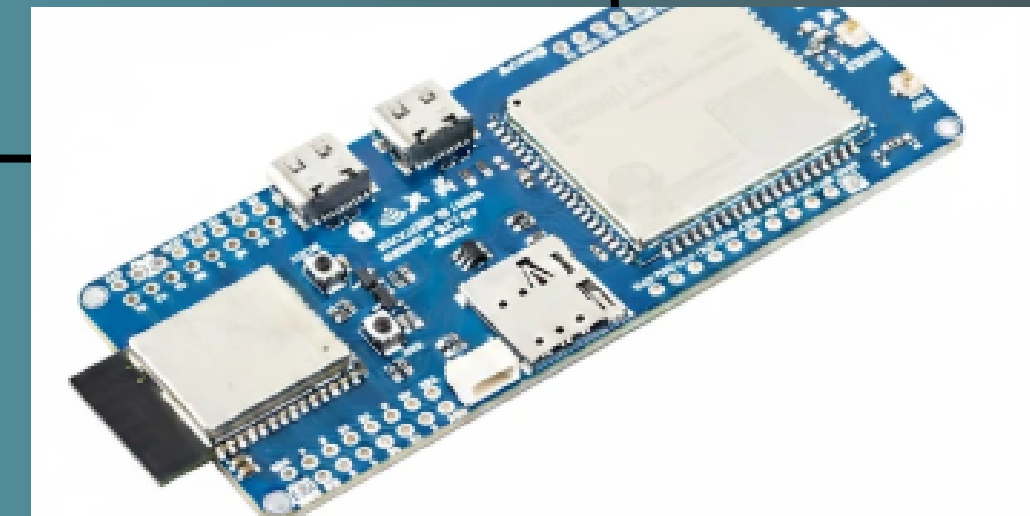


PMS5003
PM1.0 , PM 2.5 , PM10

UART 2 - RX = 42 , TX = 41

ESP32 - S3 + EC200U
4G LTE

MQTT & cloud



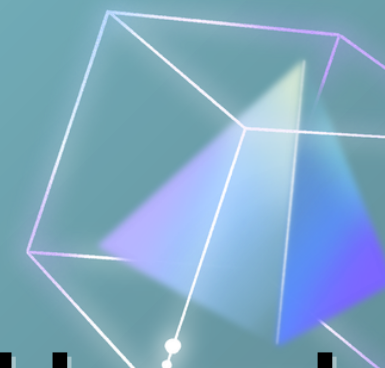
Tech Stack



- ✓ Bridge Script - Node Js
- ✓ Database → Supabase - PostgreSQL
- ✓ React + Vite — the app framework
- ✓ Tailwind CSS + Glassmorphism - the visual styling
- ✓ Deployed on Render
<https://iaqiot-1.onrender.com/>



MQTT & Cloud Pipeline



Sensor → ESP32 → EC200U → EMQX Broker → Bridge → Supabase → Dashboard

1 ESP32 -S3

Reads sensors ,
Builds JSON
payload every 20
seconds

2 EC200U

Modem
4G LTE uplink AT
command MQTT
over Airtel GPRS

3 EMQX Broker

broker.emqx.io
port 1883
Topic:
iaq/palakkad/dat
amos

4 Bridge

Receives
json
Parses it
Insert to
supabase

Bridge Script - the backend

A lightweight Node.js script that runs 24/7 on a free cloud server that acts as the glue between EMQX and Supabase.

What it does (3 steps):

1. Subscribes to EMQX broker on topic `iaq/palakkad/datamos`
2. Parses the incoming JSON payload from ESP32
3. Inserts a new row into Supabase `iaq_readings` table every 20 seconds

Database - Supabase (PostgreSQL)



Why DB?

Without database , user can only see the current data. The moment next reading comes , the current one is lost



Every Sensor Reading is permanently stored



Enables Time Series Analytics



Works in Real time and pushes to dashboard instantly via web socket



Historical Date Range filtration

```
"username" => null
"gender" => "admin"
"email" => null
"email_verified_at" => null
"password" => "$2y$10$1rmuss
"isActive" => 1
"user_role" => "Administrat
"avatar" => "assets/img/use
"remember_token" => "0dwr7S
"created_at" => "2022-01-01"
```


Front end Dashboard

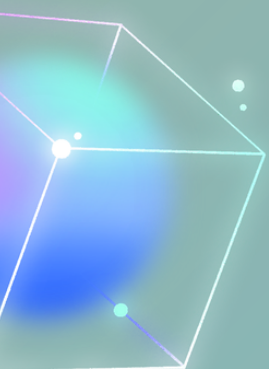
01. Supabase Realtime detects the new INSERT and immediately pushes that row to all connected clients via WebSocket without polling, no delay

02. React receives the new row and updates its state with latest reading tiles (AQI, CO₂, PM2.5 etc.) instantly show the new values

03. Recharts reads the updated state and re-renders the line chart with one new data point appended to the right — the graph grows live in real time



End to End System Flow



✓ **Sensors**

SDC4x + ENS160
+ PMS5003

✓ **ESP32 S3**

Rolling Average ,
AQI Calculation ,
Json build

✓ **EC200U 4G**

AT Commands
MQTT Publish
every
20seconds

✓ **EMQX Broker**

iaq/palakkad/dat
amos

✓ **Bridge script**

Connects MQTT
and Supabase



✓ **Supabase**

Stores all the
readings and
helpful for
analytics

✓ **Front end
Dashboard**

view the results
in interactive
dashboard

Thank You



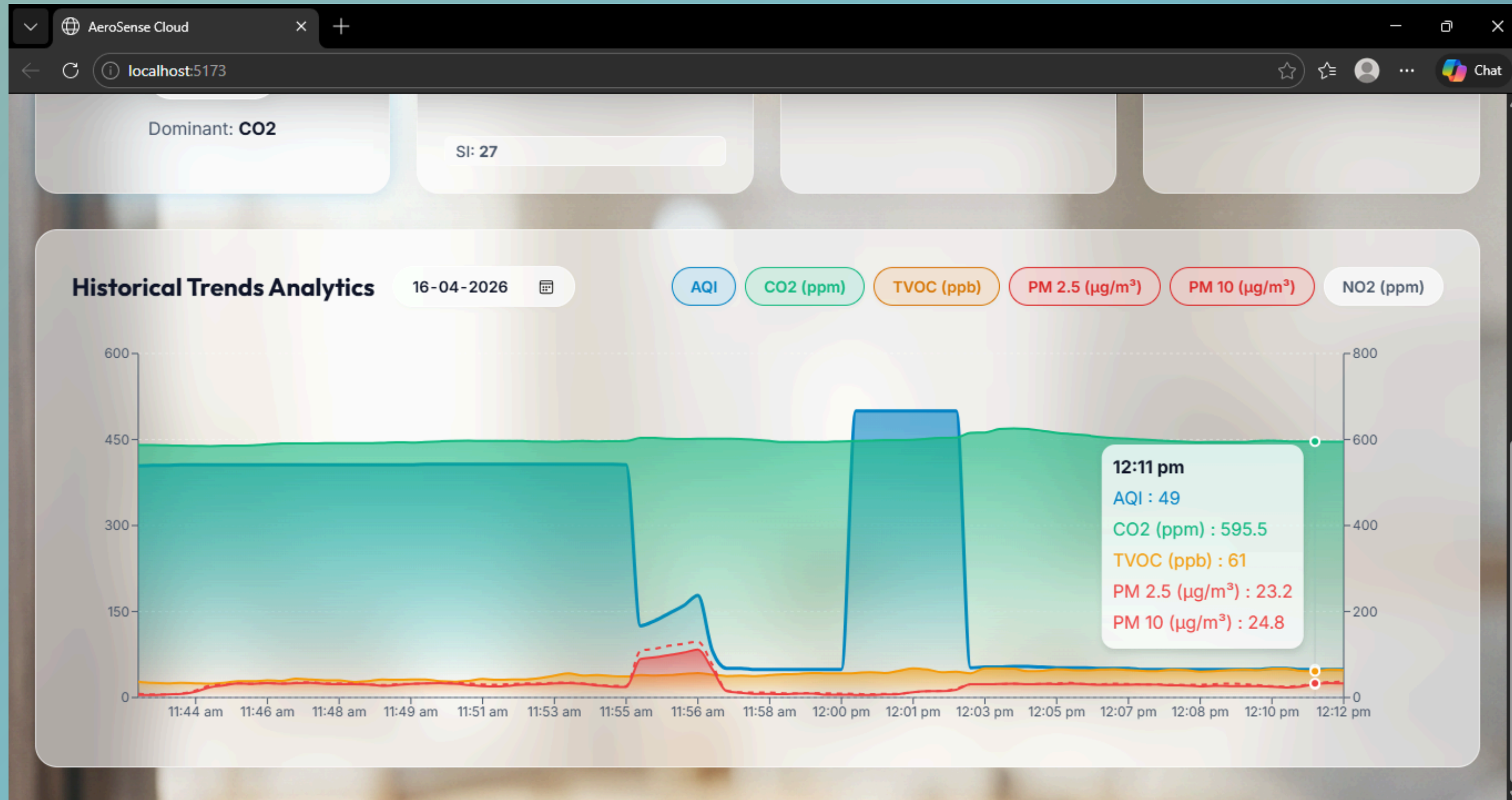
Recorded in IPTIF to ICSR



Recorded in Hostel Room :



Recorded in Common Study Room :



Recorded in C06 Lab during OELP presentation

